

Code Overview

WIFI SETUP AND SERVER

This section outlines the **WiFi connection setup** and how to implement a web server on the ESP32 to control the LED.

Working Principle of the Project

HOW IT OPERATES

The ESP32 connects to WiFi and creates a local web server. Users can access the server via an IP address in their browser. Buttons on the webpage allow users to control the LED, turning it ON or OFF easily.



Circuit Diagram Overview

ESP32 TO LED CONNECTION



Components Required for Project

ESSENTIAL PARTS FOR SMART LED BLINKER

- ESP32 board for WiFi connectivity
 - LED to indicate device status
 - 220Ω resistor to limit current flow
 - Breadboard & jumper wires for connections
 - Reliable WiFi connection for remote control
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Project Objectives

KEY GOALS FOR THE PRESENTATION

- Control LED using a web browser
 - Learn the basics of Internet of Things
 - Understand the web server in ESP32
 - Explore practical applications of ESP32 technology
 - Enhance skills in embedded systems development
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Understanding ESP32 and IoT

WHAT IS ESP32?

ESP32 is a low-cost, low-power microcontroller with integrated WiFi and Bluetooth capabilities, ideal for IoT applications and embedded systems.

WHAT IS IOT?

The Internet of Things (IoT) refers to interconnected devices that communicate over the internet, enabling remote monitoring and control of various applications.

Smart LED Blinker

